Chapter 5

Calibration Procedures

INTRODUCTION

All quality systems that address calibration require documented instructions for the calibration of inspection, measurement, and test equipment (IM&TE). Different standards have various requirements, and those are addressed in Chapter 17, "Industry-Specific Requirements," wherever they differ from the information in this chapter. Under a quality system, this is the *say what you do* portion. This refers to the need to document, in detail, how to perform laboratory activities, including calibration; this includes calibration procedures, standard operating procedures (SOPs), protocols, job aids, work instructions, work cards, and so on. Why follow formal and documented instructions or procedures? The answer is simple: To get consistent results from a calibration, one must be able to follow the same step-by-step instructions every time a calibration activity is performed.

The importance of documented procedures is included in multiple published standards. To begin with, ISO 10012-2003, Clause 6.2.1, states, "Measurement management system procedures shall be documented to the extent necessary and validated to ensure the proper implementation, their consistency of application, and the validity of measurement results. New procedures or changes to documented procedures shall be authorized and controlled. Procedures shall be current, available and provided when required."[1]

ANSI/ASQC M1-1996, Clause 4.9, states, "Documented procedures, of sufficient detail to ensure that calibrations are performed with repeatable accuracy, shall be utilized for the calibration of all ensembles."[2]

ISO 9001:2015 is focused on outputs. It is silent on the requirements for details *within* a calibration procedure other than to require that "[t]he organization shall determine and provide the resources needed to ensure valid and reliable results when monitoring or measuring is used to verify the conformity of products and services to requirements."[3]

ISO/IEC 17025:2017 is also silent on the requirements *within* a calibration procedure and is also focused on outputs. It does require a laboratory to "document its procedures to the extent necessary to ensure the consistent application of its laboratory activities and the validity of the results."[4]

The common theme across these standards is to document calibration method procedures with enough detail to ensure consistent application and valid results. So, what should be in a calibration method procedure to ensure these outputs? Some companies use original equipment manufacturer (OEM) procedures that are

10. Human factors and suitability
11. Environmental and stabilization requirements
12. Calibration reporting requirements
13. Data recording, analysis, and presentation requirements
14. Handling and storage of calibration items
15. Labeling and sealing requirements
16. Configuration and change control, and
17. Safety of personnel and equipment."[5]

What could a calibration procedure look like? Formatting of how a calibration procedure appears can vary. With the exponential growth of information on the internet, examples can be found with a simple search in a search engine. Reputable sources, such as National Metrology Institutes and government agencies, may make their calibration procedures free to access. As a start, one can follow the formats of these procedures while also ensuring that the resulting procedure contains enough information and detail that one will be able to successfully and repeatedly perform a calibration to produce valid and sound results. Your own company may already have a designated format for procedures. Calibration procedures should be among the controlled documentation as specified in the quality management system.

As with any procedure or work instruction, there should be an order to the procedure where sections can build upon one another. It is helpful to reference the list of factors from NCSL RP-3 (provided earlier in this chapter) when adding detail.

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Title and Revision Control

For example, begin with the procedure title (and, if used, the associated numbering or abbreviation identifier) and information regarding the document control revision level and status. Whether in the procedure itself or elsewhere in the document control system, the revision history of the procedure should be included. If the revision history is to go in the calibration procedure, place it where it makes the most sense for your laboratory and use of the procedures.

Purpose

A section that describes the purpose of the procedure can be helpful. The purpose can include what this procedure is used for: What are the desired results or outcomes of the procedure? What is the type of IM&TE to be calibrated? What measurement results and equipment does it apply to? Are there published standards or other defined requirements that should apply to the measurement results? It is helpful to include these points in sufficient detail to ensure consistent application and valid results.

Environmental and Stabilization Requirements

What are the applicable environmental conditions and limits in which the calibration should be performed? What, if any, are the requirements for environmental stabilization of the reference standards, IM&TE, and calibration environment?

OTHER SOPS AND COMPETENCIES

What other SOPs and/or competencies are necessary for the performance of this calibration? There should be enough information about the referenced SOPs that the reader knows and understands what they are.

SAFETY AND HANDLING CONSIDERATIONS

What are any health and safety considerations for the personnel performing the calibration? How should the equipment involved in the calibration (reference standards, supporting equipment, apparatuses, reagents, IM&TE, etc.) be handled to ensure safety and integrity of the equipment? Items to consider include, but are not limited to, safety glasses, steel-toed shoes, hardhats, electrostatic discharge (ESD) straps, gloves, lab coats, respirators, and so on. This section could list the specific details or could reference the controlled document(s) that provide these details.

Handling instructions for equipment may be found in a number of sources, such as an all-encompassing procedure, within individual procedures, in equipment user or operational manuals, in supplemental materials, or elsewhere. When handling instructions are referenced in their own document, that document should be controlled in the document control system.

Reference Standards and Apparatus

List and describe the reference standards and any supporting equipment needed (including fixtures, tools, apparatuses, safety equipment, reagents, and other items). For this equipment, what are their requirements for parameters, ranges, accuracy, uncertainty, precision, resolution, etc.? Decide what and how much of this information to include in the procedure. Some labs will include the unique identification of the reference standards to be used. Although this may be helpful to easily identify which standard to use, if the list of reference standards is limited to *specific* reference standards and that specific standard is not available, then the lab would not be conforming to its procedure unless there is a caveat, such as that "or equivalent" standard can be used. When using "or equivalent" as a caveat, it would be helpful to define what "or equivalent" means regarding performance, accuracy, or other defined characteristics of the reference standard.

Symbols and Formulas

Are there unique symbols and/or formulas for this procedure? They should be clearly defined. Formulas for calculating data should be included. For complex formulas, it is good practice to define or describe the order of operations so that each person using the procedure applies the formula in the same manner.

equipment, and published standards (such as test methods, international standards) that were not specifically used in the performance of the calibration and that are also helpful to document.

METHOD VALIDATION AND VERIFICATION

It is not enough to simply draft a calibration procedure or download one from the internet and then begin using it. Many international standards require that, at a minimum, calibration methods be appropriate for the intended use and for the laboratory to verify that by performing the method they can produce repeatable and valid results.

ISO/IEC 17025:2017 7.2 is an entire clause about *Selection, verification, and validation of methods.* In this standard, "method" is considered synonymous with the term "measurement procedure" as defined in the *VIM* (JCGM 200-2012).[7]

How does one know when to "validate" versus "verify" a method? Let us begin by defining the words *validation* and *verification*.

> ***Validation*** (*VIM* 2.45): ***verification****, where the specified requirements are adequate for intended use*.[8]
>
> ***Verification*** (*VIM* 2.44): *provision of objective evidence that a given item fulfils specified requirements.*
>
> Note 5 for Verification: *Not every verification is a* ***validation****.*[9]

At first glance, the definitions seem so similar. When does a method have to be validated rather than verified? In the case of methods that have been "published either in international, regional, or national standards, or by reputable technical organizations, or in relevant scientific texts or journals, or as specified by the manufacturer of the equipment,"[10] these methods are often considered to have been developed and validated for their intended use. In these instances, if a laboratory uses the method as intended without any modifications, then the laboratory would only need to *verify* that it can correctly and repeatedly perform the method as written and generate valid measurement results. Some ways that this verification can be substantiated is by performing the method in repeatable conditions, such as in an intra-laboratory comparison. Another way is by comparison with external labs, such as in inter-laboratory comparisons or proficiency tests. Other options are listed in ISO/IEC 17025:2017, Clause 7.7.[11]

If a laboratory develops its own method, uses a nonstandard method, or modifies or uses a standard method outside the intended scope, then the laboratory must first *validate* that the method is appropriate and adequate for use (see definition of *validation*). The validation must be "a planned activity and performed by competent personnel equipped with adequate resources."[12] In a planned activity, how the method is to be performed and the measures of validation are predetermined.

Validation is an example of applying the plan–do–check–act quality tool. The *plan* includes developing and documenting the protocol (i.e., the steps and expected results). The *do* is the execution of the plan (i.e., the validation protocol.) The *check* is comparing the results of the doing to the expected results in the plan. Based on the differences, changes may be made to the plan and the process repeated.

ISO/IEC 17025:2017, Clause 7.2.2.1, lists ways that a method can be validated, including statistical analysis, testing method robustness through variation of controlled parameters, comparison of results with other validated methods, and interlaboratory comparisons (and proficiency tests).[13]

Once a method has been validated, assessed, and proven that it is appropriate for its intended use, and that the intended use is relevant to the specified needs and requirements, then the laboratory must *verify* that it can repeatedly perform the method and generate valid results. A laboratory must verify that it can successfully perform the method before implementing the method in actual use. Whether validating or verifying, the results of these activities must be recorded and also include "a statement on the validity of the method, detailing its fitness for the intended use."[14]

SUMMARY

Remember, the purpose of the calibration procedure is to provide the sufficiently detailed information and directions necessary to ensure proper and consistent application so that the measurement results are repeatable and valid by those competent and authorized to perform the procedure. The calibration procedure is the document of *what is to be done*. The calibration results are a record of *what was done*. Laboratories shall verify and/or validate methods to prove that they have competent people and the appropriate equipment and can consistently generate required outcomes of the calibration and consistently produce valid data.